**QUAID-I-AZAM UNIVERSITY ISLAMABAD**

**DEPARTMENT OF COMPUTER SCIENCE**




# DETECTION OF CLICKBAIT THROUGH MACHINE LEARNING

## SOFTWARE PROJECT TECHNICAL DOCUMENTATION


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**Submitted to:**

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# *Abstract*

## *Detection of clickbait through Machine Learning*

Due to the fast development of social media and information websites, misleading and exaggerated news titles, have become increasingly more commonplace. These misleading information titles goal to seize interest of the reader and maximize the quantity of clicks. However, they'll mislead the reader and create false expectations. In this assignment, the development of a **Clickbait Detection System** based totally on system getting to know may be discussed. The system might be capable to analyze news titles or small information articles to decide whether they're clickbait or valid information.

To increase the **Clickbait Detection System, Natural Language Processing (NLP)** strategies might be hired to procedure the input information, extract essential features, and train a class version to become aware of clickbait, The system might be capable of be given news titles or small news articles as input and offer an immediate prediction with a self assurance level. The machine will also be capable of pick out the vital phrases or terms that inspired the prediction to help the person better recognize the purpose behind the prediction. In addition to this , the system can also be able to maintain a history of the predictions and offer accuracy, precision, and take into account metrices to determine the performance of the type version.

By integrating gadget mastering with the prediction mechanisms, the proposed Clickbait Detection System may be an sensible machine to become aware of misleading news titles. The proposed device could be beneficial for customers to critically examine the virtual content they eat, and the overall recognition of deceptive news titles can be elevated

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# Chapter No. 01

# Software Project Management Plan

# 1 Introduction

## Project Overview

- **Purpose:** The purpose of this is to develop a machine learning based software system that will automatically detects **Clickbait** content from a **news headlines** or **short text**. The aim of the system is to help users to identify the **wrong and dramatic headlines** by providing the accurate predication, with easy explanation to make online content clear and trustworthy.

- Objectives:
    - Design and implement this system for which machine learning will be capable of classifying text as:
        1. Clickbait
        2. Non-Clickbait

Perform text preprocessing and feature extraction to convert raw text into a machine readable format.

- Machine studying version will teach using a labeled dataset containing clickbait and non-clickbait headlines.
- A consumer-friendly software program is increase for interface that lets in users to:
    1. For evaluation input brief text or news headlines.
    2. View the prediction end result(Clickbait/Non-Clickbait).
    3.See the essential words that stimulated the prediction
- Dataset shop the detection results so customers can:
    1. History of previous Clickbait detection may be view.
    2. Analyze Clickbait pattern over the years.
- Allow uploading new information to keep the detection model updated.

Scope:

- Building a Software primarily based device for clickbait detection the usage of a device gaining knowledge.
- The device cleans text, extracts features the usage of TF-IDF , teach the model ,and make the predictions.
- Design an easy interface to enter text and look at explanantions.

- The predominant consciousness of the machine is on English language headlines and brief text content.
- The purpose of task is to detect clickbait with machine studying model which might be comprehensible and adaptable.
- The system makes use of device mastering to assist customers recognize and understand clickbait headlines actually.

## 1.2 Project Deliverables

Project deliverable includes:

1) **Technical Documentation:**
    a) Software Project Management Plan
    b) Software Requirement Specification
    c) Software Design Description
    **d)** Software Test Documentation **Web Application**
2) **Clickbait detection system**
    a. **Interactive text analysis interface**
    To submit headlines and short text and view the classification result.
    b. **Explainable Decision Module**
    Helping users understand the result by highlighting important words.
    c. **Prediction Record Management Module**
    To store old headlines and see their result again.
    d. **Adaptive Learning Module**
    The model can be improved by using new data
3) **Machine learning and AI components**
    a. **Text Classification Engine**
        i. Logistic Regression / Naïve Bayes Classifier
    b. **Text Feature Extraction**
        i. TF-IDF vectorizer
        ii. N-gram feature representation
    c. **Explainable AI Component**
        i. Influential word identification module
        ii. Feature weight analysis
    d. **Model Evaluation & adaptive Module**
        i. Accuracy, Precision and recall measurement
        **ii.** Model retraining mechanism

# 2 Project Organization

## 2.1 Software Process Model

For the development of Detection of clickbait through machine learning, an Agile Software Development Model is chosen.

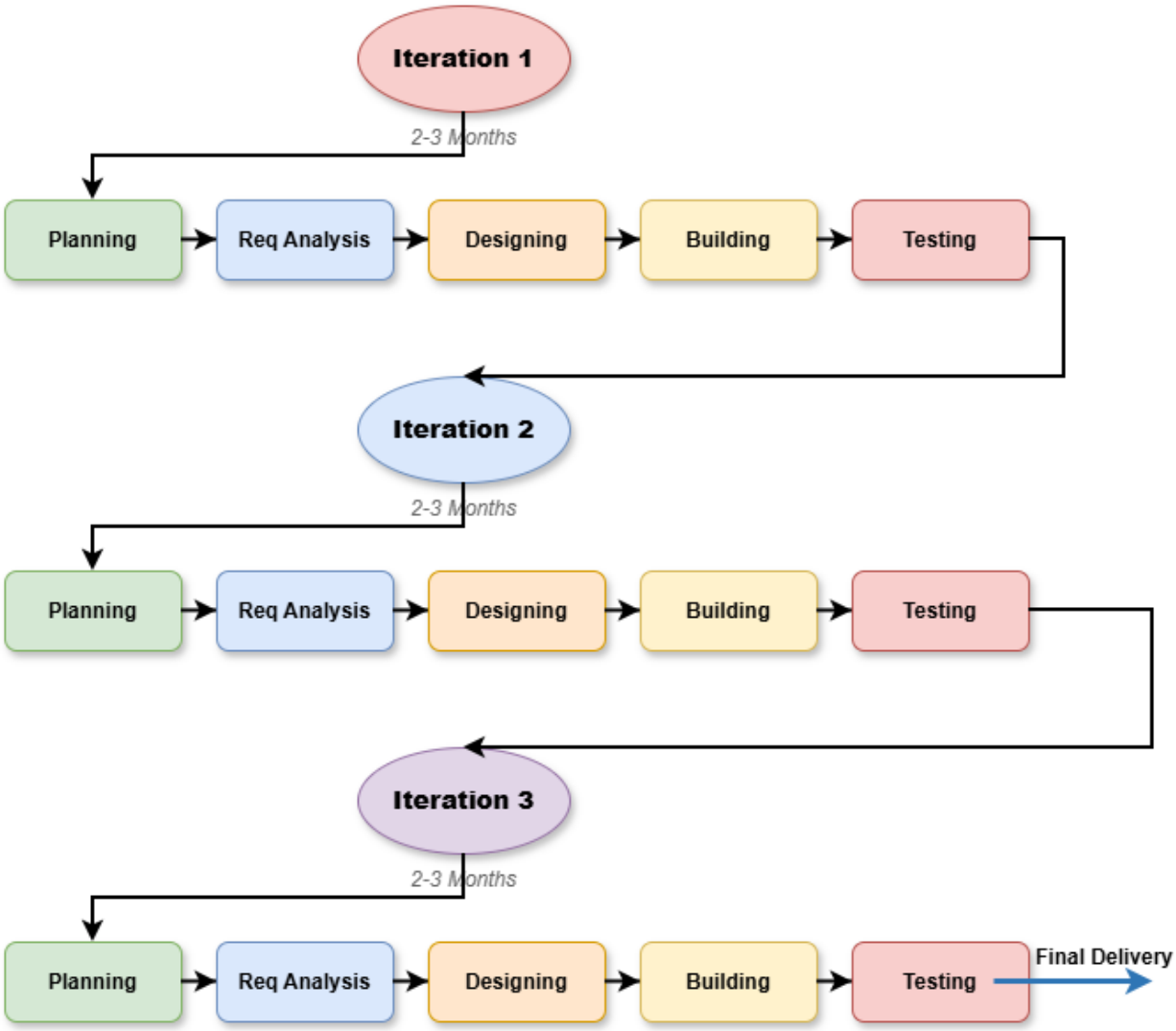



*Figure 1: Agile Software Development Model*

Agile methodologies are iterative and incremental , which mean they are known for dividing the project into smaller parts and adjusting to changing requirements. It prioritizes flexibility, collaboration, and customer satisfaction.

## 2.2 Roles and Responsibilities

1. **Student ( Faizan Ahmad, Zohaib Zafar ):**

a. System Analyst
b. Software Architect
c. Machine Learning Model Designer
d. Dataset Curator and Preprocessing Analyst
e. UI / UX Designer
f. Front-end Developer
g. Software Developer
h. Explainable AI Analyst
i. Quality Assurance Engineer

2. **Project Supervisor Faizan:**
    a. Mentor
    b. Technical Reviewer
    c. Academic and research guidance

## 2.3 Tools and Techniques

| Tools / Technique | Purpose |
|---|---|
| Project Libre | For creating Project Timeline. |
| Microsoft Word | For creating technical project documentation. |
| Draw.io | For designing UML diagrams. |
| Public Clickbait Dataset | Used for training and evaluating the detection model. |
| Flask | It handles user requests and sends them to the model. |
| Natural Language Processing (NLP) | Used for checking and processing headline textual content. |
| Postman | For API testing. |
| Python | Used to create core system and machine learning components. |
| Scikit-learn | Used to build and evaluate clickbait detection models. |
| Vs Code | A light-weight code editor for NodeJS, Express.JS programming. |
| TF-IDF Vectorizer | Used to change text data into numerical features. |
| Feature Importance Analysis | Used to discover influential phrases contributing to prediction end result. |
| Git GitHub | For version control and collaboration. |
| SQLite / CSV-based Storage | Used for storing prediction records and analysis history. |
| Web-based interface | For consumer interaction and end result visualization. |
| HTML, CSS and javaScript | It is used to create the screen in which customers enter headlines and spot results. |
| Figma | For creating UI design. |
| Word Tokenization | This is used to break text into words. |
| Logistic Regression, Naïve Bayes | Used to classify headlines as clickbait or non-clickbait. |
| Accuracy, Precision, Recall, F1-score | This helps in testing how good the model is. |
| Feature weight analysis | Help to locate vital phrases behind the prediction. |
| Local Web Server | It will help to run and reveal the device in the course of evalution. |